



Selected Topics in Cryptography

Session 07: Signature with schoolbook RSA and CRT

April 21, 2026

Develop the following exercises in pairs. You can use C/C++, Java or Python to develop your source code, but please use only one of these languages to develop all the exercises. You can use a cryptographic library but only for specific purposes described in each exercise. Also you can use a library to do arithmetic with big integers.

1. Programming exercises

1. Design and implement a function to generate the pair of keys for RSA with CRT. Please consider the following requirements:
 - You must use a function in the cryptographic library of your choice to generate random numbers .
 - The RSA module n must have 1024 bits. You can use a function on a cryptographic library to generate each prime p and q . Choose the correct length of the primes.
 - You can use a function in a library to compute a multiplicative inverse and/or calculate the greatest common divisor.
 - Your function must store the private key in base 64 in a text file. The public key must be stored, also in base 64 in a different text file.
2. Design a function to sign a message M stored in a textfile using SHA2 and RSA with CRT. Your function must receive a private key and the name of the textfile. Use SHA2 obtain the hash h of M , then use the private key of RSA and h to obtain the signature s . Print s in base64.
3. Design a function to verify the signature of a message M stored in a textfile using SHA2 and RSA. Your function must receive a public key e , the name of the textfile, and the signature s in base64. Use SHA2 obtain a hash h' of M . Then do the operation $h = s^e \bmod n$ to recover h . Compare h and h' and return true if the signature is valid.
4. Do a program to test the previous functions, with different files. One of the students in the team must generate a key pair for RSA, and sign a message. Then the student must share his/her public key, the message and the signature to his/her partner. The partner must verify the signature.

2. Products

Every student must write her/his own report including the following information:

1. Your personal information, date of the lab session and the topic that we are studying in this lab session.
2. The source code used to implement the points 1.1, 1.2 and 1.3. Please include a brief paragraph explaining each function.
3. Screenshots of your programs running.